Engineering Calculus
EMAT101L

Lecture 2

School of Engineering Science & Technology,

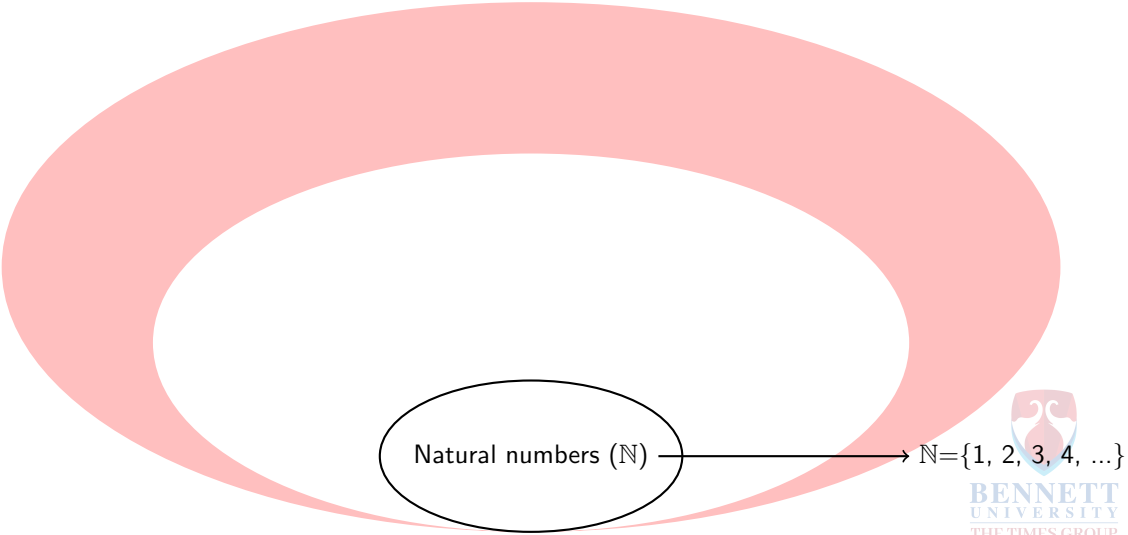
Bennett University

September 8, 2023

- The Real number system,
- Archimedean property,
- Sequence and its convergence and divergence,
- Monotone sequences,
- Cauchy Criterion,
- Bolzano-Weierstrass Theorem,
- Limit inferior and limit superior.
- Infinite Series and its convergence,
- Alternating series,
- Limit and continuity,
- Existence of maxima & minima,
- Intermediate Value Property.

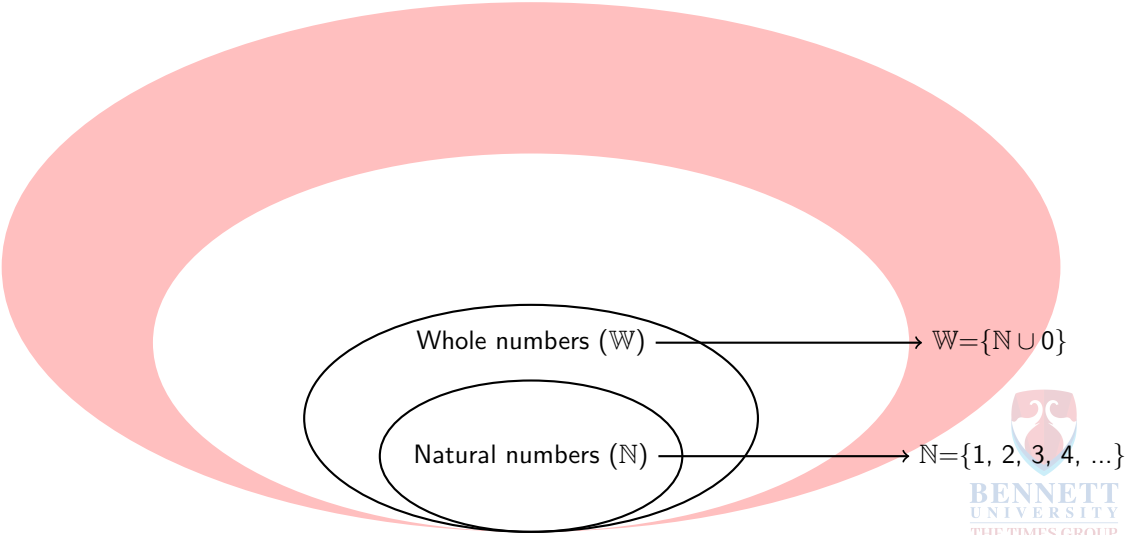
BENNETT UNIVERSITY
THE TIMES GROUP

The real number system



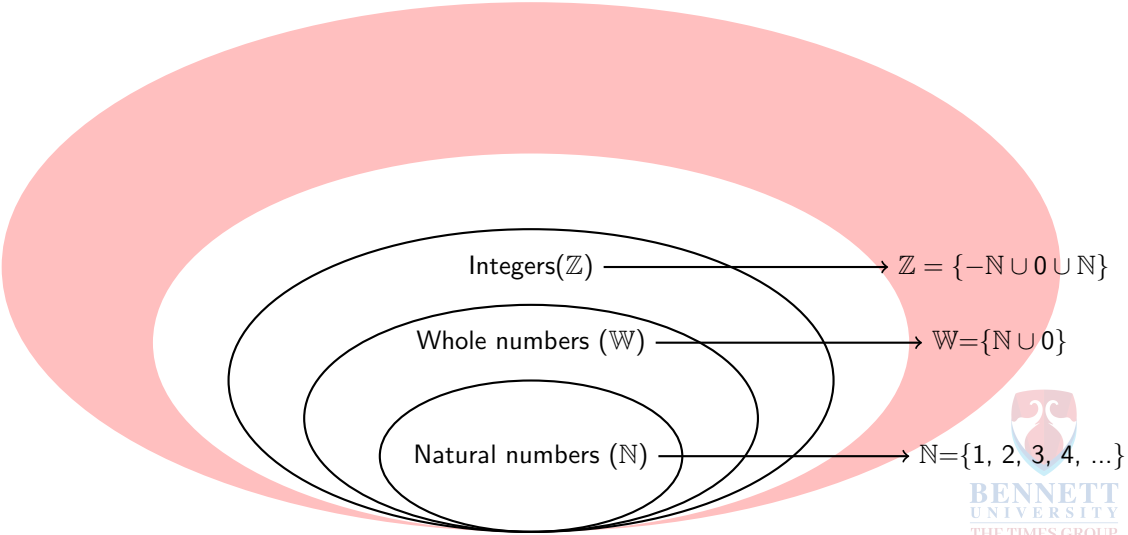
BENNETT UNIVERSITY
THE TIMES GROUP

The real number system



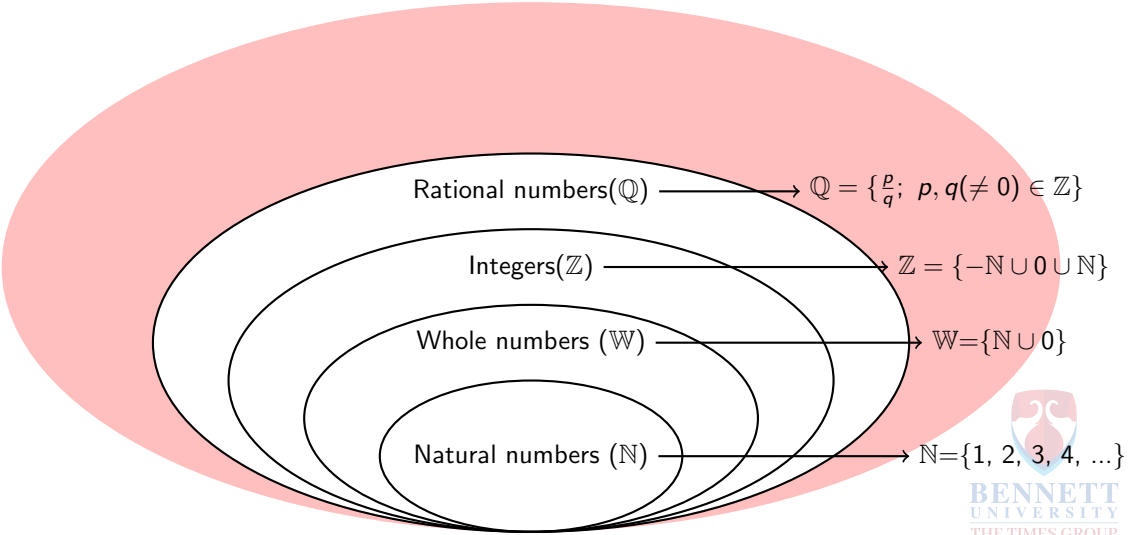
BENNETT UNIVERSITY
THE TIMES GROUP

The real number system



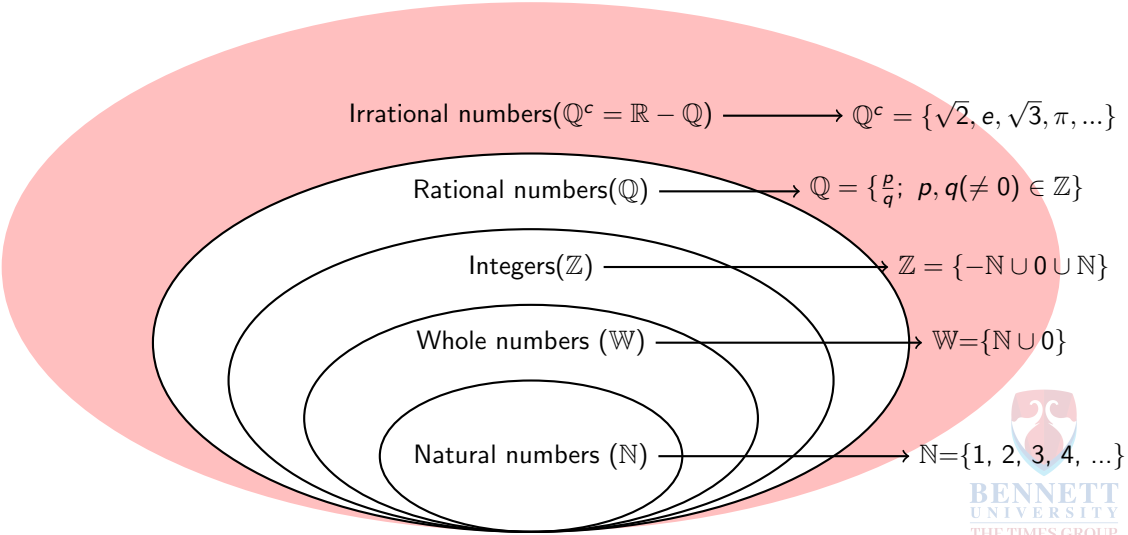
BENNETT UNIVERSITY
THE TIMES GROUP

The real number system



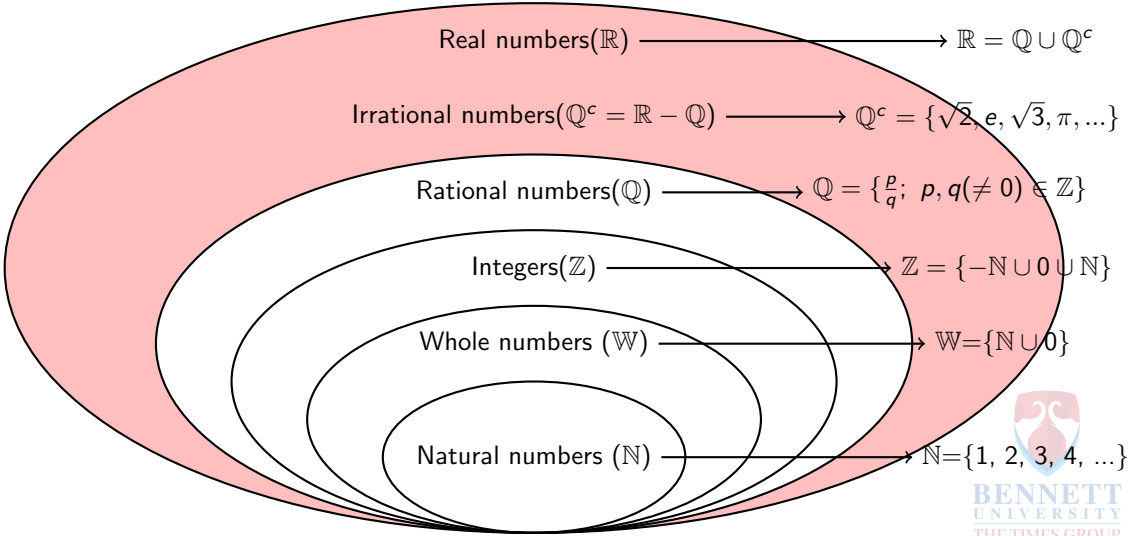
BENNETT UNIVERSITY
THE TIMES GROUP

The real number system



BENNETT UNIVERSITY
THE TIMES GROUP

The real number system



BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. π

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. π =3.14159265358979323846264338327950288419 7...

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. π =3.14159265358979323846264338327950288419 7...
4. 22/7

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. $\pi = 3.14159265358979323846264338327950288419 7\ldots$
4. $22/7$
5. $\sqrt{5}$

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. π =3.14159265358979323846264338327950288419 7...
4. 22/7
5. $\sqrt{5}$
6. e (Euler's number)

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. π =3.14159265358979323846264338327950288419 7...
4. 22/7
5. $\sqrt{5}$
6. e (Euler's number) =2.71828182845904523536028747135 27...

Archimedean Property

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

1. 2
2. 2.5
3. π =3.14159265358979323846264338327950288419 7...
4. 22/7
5. $\sqrt{5}$
6. e (Euler's number) =2.71828182845904523536028747135 27...

Archimedean Property

If x and y are any two positive real numbers with $y < x$, then there exists a positive integer n such that $ny > x$.

BENNETT UNIVERSITY
THE TIMES GROUP

The distance

The most convenient and familiar way to represent real number system is the real line $\mathbb{R}$

BENNETT UNIVERSITY
THE TIMES GROUP

The distance

The most convenient and familiar way to represent real number system is the real line $\mathbb{R}$

Real line

Each number 'a' on the line represents the distance from the origin, i.e., 0.

BENNETT UNIVERSITY
THE TIMES GROUP

The distance

The most convenient and familiar way to represent real number system is the real line $\mathbb{R}$

Real line

Each number 'a' on the line represents the distance from the origin, i.e., 0.

Problem 1 What is the distance between points 0.2 and 0.8.

BENNETT UNIVERSITY
THE TIMES GROUP

The distance

The most convenient and familiar way to represent real number system is the real line $\mathbb{R}$

Real line

Each number 'a' on the line represents the distance from the origin, i.e., 0.

Problem 1 What is the distance between points 0.2 and 0.8.

Problem 2 : What is the distance between points 0.2 and -0.8.

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

Half closed/open interval

$$[a, b) = \{x \in \mathbb{R} : \ a \leq x < b, \ a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : \ a < x \leq b, \ a, b \in \mathbb{R}\}$$

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

Half closed/open interval

$$[a, b) = \{x \in \mathbb{R} : \ a \leq x < b, \ a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : \ a < x \leq b, \ a, b \in \mathbb{R}\}$$

Length of intervals in all above cases is $b - a$. If $a = b$ then
Q1: $(a, a) =$

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

Half closed/open interval

$$[a, b) = \{x \in \mathbb{R} : \ a \leq x < b, \ a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : \ a < x \leq b, \ a, b \in \mathbb{R}\}$$

Length of intervals in all above cases is $b - a$. If $a = b$ then
Q1: $(a, a) = \phi$

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

Half closed/open interval

$$[a, b) = \{x \in \mathbb{R} : \ a \leq x < b, \ a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : \ a < x \leq b, \ a, b \in \mathbb{R}\}$$

Length of intervals in all above cases is $b - a$. If $a = b$ then

Q1: $(a, a) = \phi$ Q2: $[a, a] =$

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

Half closed/open interval

$$[a, b) = \{x \in \mathbb{R} : \ a \leq x < b, \ a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : \ a < x \leq b, \ a, b \in \mathbb{R}\}$$

Length of intervals in all above cases is $b - a$. If $a = b$ then

Q1: $(a, a) = \phi$ Q2: $[a, a] = \{a\}$.

BENNETT UNIVERSITY
THE TIMES GROUP

Intervals

Open interval

$$(a, b) = \{x \in \mathbb{R} : \ a < x < b, \ a, b \in \mathbb{R}\}$$

Close interval

$$[a, b] = \{x \in \mathbb{R} : \ a \leq x \leq b, \ a, b \in \mathbb{R}\}$$

Half closed/open interval

$$[a, b) = \{x \in \mathbb{R} : \ a \leq x < b, \ a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : \ a < x \leq b, \ a, b \in \mathbb{R}\}$$

Length of intervals in all above cases is $b - a$. If $a = b$ then

Q1: $(a, a) = \phi$ Q2: $[a, a] = \{a\}$. $\mathbb{R} = (-\infty, \infty)$

Infinite open interval

$$(a, \infty) = \{x \in \mathbb{R} : \ x > a, \ a \in \mathbb{R}\}$$
$$(-\infty, b) = \{x \in \mathbb{R} : \ x < b, b \in \mathbb{R}\}$$

Infinite closed interval

$$[a, \infty) = \{x \in \mathbb{R} : \ x \geq a, \ a \in \mathbb{R}\}$$
$$(-\infty, b] = \{x \in \mathbb{R} : \ x \leq b, b \in \mathbb{R}\}$$

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.
Q2: If $S = [1, 2)$, what is $max(S)$ and $min(S)$?

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.
Q2: If $S = [1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, 1.

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.
Q2: If $S = [1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, 1.
Q3: If $S = (1, 2)$, what is $max(S)$ and $min(S)$?

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.
Q2: If $S = [1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, 1.
Q3: If $S = (1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, do not exist.

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.
Q2: If $S = [1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, 1.
Q3: If $S = (1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, do not exist.
Q4: If $S = \{1, 1/2, 1/3, 1/4, ...\}$, what is $max(S)$ and $min(S)$?

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M, of the set S (if exists) is called the maximum of the S.

Minimum of a set

The smallest element, m, of the set S (if exists) is called the minimum of the S.

Problems:
Q1: If $S = [1, 2]$, what is $max(S)$ and $min(S)$?
Ans: 2, 1.
Q2: If $S = [1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, 1.
Q3: If $S = (1, 2)$, what is $max(S)$ and $min(S)$?
Ans: Do not exist, do not exist.
Q4: If $S = \{1, 1/2, 1/3, 1/4, ...\}$, what is $max(S)$ and $min(S)$?
Ans: 1, There is no minimum as for all $1/n \in S$ the number $1/(n+1)$ is in the set and smaller..

BENNETT UNIVERSITY
TIMES GROUP

Bounds

Bounded above

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded above** *if there is* an element $K \in \mathbb{R}$ such that $x \leq K$ for all $x \in S$. The number K is said to be **upper bound** of S.

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Bounded above

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded above** *if there is* an element $K \in \mathbb{R}$ such that $x \leq K$ for all $x \in S$. The number K is said to be **upper bound** of S.

Bounded below

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded below** *if there is* an element $k \in \mathbb{R}$ such that $x \geq k$ for all $x \in S$. The number k is said to be **lower bound** of S.

BENNETT UNIVERSITY
THE TIMES GROUP

Bounds

Bounded above

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded above** *if there is* an element $K \in \mathbb{R}$ such that $x \leq K$ for all $x \in S$. The number K is said to be **upper bound** of S.

Bounded below

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded below** *if there is* an element $k \in \mathbb{R}$ such that $x \geq k$ for all $x \in S$. The number k is said to be **lower bound** of S.

Bounded set

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded** if it is bounded above as well as bounded below.

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above,

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below,

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded,

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
- Upper bounds of S

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
- Upper bounds of $S = [2, \infty)$

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
• Upper bounds of $S = [2, \infty)$ and • Lower bounds of S

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
• Upper bounds of $S = [2, \infty)$ and • Lower bounds of $S = (-\infty, 1]$.

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
- Upper bounds of $S = [2, \infty)$ and • Lower bounds of $S = (-\infty, 1]$.
- S is bounded.

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
• Upper bounds of $S = [2, \infty)$ and • Lower bounds of $S = (-\infty, 1]$.
• S is bounded. • All results hold same for $S = [1, 2]$.

BENNETT UNIVERSITY
THE TIMES GROUP

Examples

Ex: $S = \{x \in (R) : x < 2\}$. Then S is
(a) Bounded above, (b) Bounded below, (c) Bounded, (d) Unbounded.
Answer: (a) and (d).
In fact, every number greater then 2 is also an upper bound of S.

Ex: If $S = (1, 2)$, then
• Upper bounds of $S = [2, \infty)$ and • Lower bounds of $S = (-\infty, 1]$.
• S is bounded. • All results hold same for $S = [1, 2]$.

Ex: Find the bounds of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$.

BENNETT UNIVERSITY
THE TIMES GROUP